



Arvind Kumar Yadav
Computer Science & Engineering
Indian Institute of Technology Bombay

B.Tech.
Gender: Male
DOB: (

Examination	University	Institute	Year	CPI / %
Graduation	IIT Bombay	IIT Bombay	2026	
Intermediate	HSC	Sunbeam Academy Samneghat Lanka Varanasi UP	2022	
Matriculation	HSC	Sunbeam Academy Samneghat Lanka Varanasi UP	2020	

Pursued a **minor** in **Mathematics**.

(2025)

SCHOLASTIC ACHIEVEMENTS

- Secured **All India Rank 192** and ranked **4th Zonal Rank** in JEE Advanced amongst 1,55,538 candidates. (2022)
- Ranked among the **National top 1%** of candidates who appeared for IOQP Part I and IOQA Part I. (2021-22)
- Achieved **All India Rank** of **198** in the *Kishore Vaigyanik Protsahan Yojana (KVPY SX)*, and **All India Rank** of **397** in the *KVPY SA* organized by Indian Institute Of Science. (2021-22,2020-21)
- Cleared the **IOQM** (Indian Olympiad Qualifier in Mathematics) examination, organized by MTAI. (2020-21)
- Secured **All India Rank 424** in Joint Entrance Examination Mains in a pool of 9,05,590 candidates. (2021)

KEY PROJECTS

Optimising libguestFS | internship

Summer 2025

MSDP malware scan

Cohesity

- Reverse-engineered the **open-source** virtualization library libguestfs, performing root cause analysis to identify performance **bottlenecks** in **listing files** from large directories in **locally mounted VMDK images**.
- Proposed a fix for a critical bug that, in testing, **reduced** file listing time for **3.9M** files from **6 hr** to **18 min**.
- Designed a **theoretical** parallel IO architecture to address **read bottlenecks** and **enable** concurrent data accesses.

Sanitising Lossy Public Key Encryptions | RnD

Spring 2025

Guide: Prof. Chethan Kamath

IIT Bombay

- Explored **advanced cryptographic** concepts by **studying research papers** on sanitization of public key encryption and on objects like lossy public key encryption, **dual-mode commitments**, and **batch proofs**.
- Investigated the **open problem** of removing decryption errors (**sanitizing**) lossy public-key encryption by pursuing **both constructive** approaches for a sanitizing scheme and **negative** results showing potential impossibility.

Blockchain P2P Network Simulation | Course Project

Spring 2025

Guide: Prof. Vinay Ribeiro | course : Introduction to Blockchains, Cryptocurrencies & Smart Contracts IIT B

- Developed a **discrete-event simulator** for a **P2P cryptocurrency** network with **exponentially** distributed **latencies** to model the Bitcoin blockchain, analyzing how **forks** correlate with different parameters.
- Simulated blockchain attacks by implementing **selfish mining** and **eclipse attacks**, studying blockchain behavior under varying malicious node counts, adversarial hashing power, network delays, and honest miner populations.

Lunar Element Detection | Krittika Computational Astronomy Projects

Winter 2024

Krittika

IIT Bombay

- Developed a Python pipeline to organize **FITS** files from the readings of the **Chandrayaan-2 Large Area Soft x-ray Spectrometer (CLASS)** for **June 2023** into Day and Night side based on the **SOLARANG** values.
- Generated a background model using the spectras from Night side to estimate the **ambient X-ray fluorescence**.
- Utilized **Gaussian curve-fitting** around the known **K_{α} lines** with the **scipy** library to detect elements like Silicon, Aluminum, and Magnesium on the different lunar coordinates.

SCLP Compiler | Course Project

Spring 2025

Guide: Prof. Uday Khedker | course : Implementation of Programming Languages Laboratory IIT Bombay

- Implemented **lexical analysis** and **parsing** for a C-like language using **Lex** for tokenization and **Bison** for **grammar-based** AST generation, applying structured grammar design and syntax-directed translation.
- Built a **full compiler** in C for the language, generating **Three-Address Code (TAC)**, **Register Transfer Language (RTL)**, and **ASM** code by leveraging object-oriented principles in C.

OTHER PROJECTS

Cybergaurd | Season Of Code

Summer 2024

Web and Coding Club(WnCC)

IIT Bombay

- Explored the basic **networking** concepts, **OSI** and **TCP/IP** models, firewalls, **DDoS** and **SQL injections**.
- Used the **Scapy** library of Python to send and receive different packets, captured them using **Wireshark**.
- Trained models on sample network data to detect anomaly and achieved more than **97%** test accuracy using NN.

Introduction to Machine Learning | Learner Space

Summer 2024

Web and Coding Club(WnCC)

IIT Bombay

- Explored the basic machine learning concepts, including linear, logistic **regressions**, and **deep learning** models.
- Explored the **YOLOv8** algorithm and **trained** a YOLOv8 model to **detect** different parts of the car accurately.
- Developed a **facial recognition** system using a **Siamese Neural Network** in **TensorFlow** to perform **one-shot** image recognition, enabling verification of whether two images represent the same person.

Algorithmic Trader | Course Project

Autumn 2023

Guide: Prof. Ashutosh Kumar Gupta | course: Data Structures and Algorithms Lab

IIT Bombay

- Developed and tested trading algorithm for buy low-sell high and detecting **arbitrage** within the order book.
- Simulated a **market** and formulated strategies to increase profit by exploiting arbitrage and moving median.

Musical Chord Encoder and Assembly | Course Project

Autumn 2023

Guide: Prof. Biswabandan Panda | course: Digital Logic Design and Computer Architecture Lab

IIT Bombay

- Developed a **VHDL**-based musical chord encoder circuit to process note inputs and determine chord types, including major, minor, suspended, and dominant sevenths while employing a **synchronized buffer mechanism**.
- Implemented heap sort in **x86**, optimized query processing over unsorted data using merge sort and binary search in **MIPS ISA** and reverse-engineered binary executables by inspecting their **x86** assemblies using **objdump**.

POSITIONS OF RESPONSIBILITY

Teaching Assistant

Spring 2023

Linear Algebra and Differential Equations(MA 110)

IIT Bombay

- Conducted weekly **tutorial sessions** for a batch of **+30 first-year** students to solve selected problems and efficiently managed students' queries during quizzes and semester exams by collaborating with the instructors.

Project Mentor

Summer 2025

Summer Of Science(SoS) | Maths and Physics Club

IIT Bombay

- Mentored students in a Summer of Science for a reading project on **blockchains**, **providing** curated learning resources, **clarifying** technical doubts, and **guiding** project planning to ensure steady progress.

Teaching Assistant

Autumn 2025(ongoing)

Course: Automata Theory | Instructor: Prof. G Sivakumar

IIT Bombay

- Working with the instructor and fellow TAs to **design** assignments and evaluate coursework for **170+** students.

TECHNICAL SKILLS

Python Libraries

Matplotlib, NumPy, Pandas, PyTorch, scikit-learn, TensorFlow.

Programming Languages

C++, C, Bash, Sed, Awk, Python, x86.

Development

HTML, CSS, Javascript, Git, L^AT_EX, React.

Tools

Wireshark, Make, MIPS.

RELEVANT COURSES UNDERTAKEN

- **Mathematics:** Real Analysis, Complex Analysis, Linear Algebra, General Topology, Differential Equations, Calculus I and II.
- **Computer Science and Engineering:** Design and Analysis of Algorithms, Theoretical Foundations of Cryptography, Web Security, Data Analysis and Interpretation, Logic and Theory of Computation, Introduction to Blockchains, Cryptocurrencies and Smart Contracts, Computer Networks, Operating Systems, Software Systems Lab, Digital Logic Design and Computer Architecture, Discrete Structures, Data Structures & Algorithms.

EXTRACURRICULAR

- Participated in a **Market Simulation** conducted by **Citadel & Citadel Securities** on **AmplifyME**. (2024)
- Completed a **reading project** on *Foundations of Astronomy & Astrophysics* through *Learner space* (2023)
- Achieved **4th place** as a member of the H9 Kho-kho team in the **Kho-Kho General Championship**. (2023)
- Completed a year-long program in **Kho-Kho** under the National Sports Organization, IIT Bombay. (2022-23)
- Completed an intensive **8-week DSA bootcamp** organized by WnCC in collaboration with **GfG**. (2025)
- Secured **runner-up** position as part of a 9-member team in the CSE Football Tournament. (2025)